Revista Brasileira de Recursos Hídricos
Brazilian Journal of Water Resources
Versão On-line ISSN 2318-0331
RBRH, Porto Alegre, v. 30, e43, 2025
Scientific/Technical Article

https://doi.org/10.1590/2318-0331.302520250025

Emergency forecasts for water levels in the Guaíba River during the 2024 unprecedented flood in South Brazil

Previsões emergenciais de níveis d'água no rio Guaíba durante a cheia sem precedentes de 2024 no Sul do Brasil

Matheus Sampaio Medeiros[1], Fernando Mainardi Fan[1] & Rodrigo Cauduro Dias de Paiva[1]

[1]Universidade Federal do Rio Grande do Sul, Porto Alegre, RS, Brasil
E-mails: matheus.sampaio@ufrgs.br (MSM), fernando.fan@ufrgs.br (FMF), rodrigo.paiva@ufrgs.br (RCDP)

Received: February 15, 2025 - Revised: July 01, 2025 - Accepted: July 24, 2025

ABSTRACT

In May 2024, the state of Rio Grande do Sul, in southern Brazil, experienced an unprecedented flood that severely impacted Porto Alegre and its metropolitan region. Water levels in the Guaíba River exceeded all historical records, reaching 5.37 meters on May 5, 2024, surpassing the previous peak of 4.76 meters recorded in 1941. In response to this extreme event, a team from the Hydraulic Research Institute at the Federal University of Rio Grande do Sul (IPH/UFRGS), including the authors of this study, mobilized efforts to perform daily operational forecasts of Guaíba water levels during the event. Despite the challenges, the forecasting system successfully identified critical moments of the flooding: flood levels exceeding 5 meters with a 3-day operational lead time; non-occurrence of levels above the 6 meters levees top; second peak above 5 meters with 8-day lead time; prolonged duration of the flooding around 30 days. The accuracy of the forecasts were crucial for mitigation measures such as preventive evacuations and emergency actions. This study highlights technical and operational aspects of the forecasts and discusses the importance of an integrated approach to managing extreme events.

Keywords: Hydrological forecasting; Unprecedented flood forecasting; Extreme events.

RESUMO

Em maio de 2024, o estado do Rio Grande do Sul, no sul do Brasil, vivenciou uma cheia sem precedentes que impactou severamente Porto Alegre e sua região metropolitana. Os níveis d'água no rio Guaíba ultrapassaram todos os registros históricos anteriores, atingindo 5,37 metros no dia 5 de maio de 2024, superando o pico anterior de 4,76 metros registrado em 1941. Em resposta a esse evento extremo, uma equipe do Instituto de Pesquisas Hidráulicas da Universidade Federal do Rio Grande do Sul (IPH/UFRGS), incluindo os autores deste estudo, mobilizou esforços para realizar previsões operacionais diárias dos níveis do Guaíba, de forma voluntária, durante o evento. Apesar dos desafios, o sistema de previsão identificou com sucesso momentos críticos da inundação com certa antecedência operacional, como: níveis superiores a 5 metros com antecedência operacional de 3 dias; a não ocorrência de níveis acima do sistema de proteção de 6 metros; um segundo pico acima de 5 metros com antecedência de 8 dias; e a longa duração da inundação, em torno de 30 dias. A acurácia das previsões foi fundamental para medidas mitigatórias como evacuações preventivas e ações emergenciais ao longo do evento. Este estudo destaca aspectos técnicos e operacionais das previsões e discute a importância de uma abordagem integrada para a gestão de eventos extremos.

Palavras-chave: Previsão hidrológica; Previsão de cheias sem precedentes; Eventos extremos.

This is an Open Access article distributed under the terms of the Creative Commons Attribution license (https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

INTRODUCTION

The establishment of monitoring centers and the development of tools for early forecasting and tracking of extreme events have gained prominence worldwide (Alfieri et al., 2013; Pappenberger et al., 2011; Pechlivanidis et al., 2025; Zhang et al., 2025). Improving these forecasting systems is essential for mitigating extreme events, such as floods, and the use of technology must be accompanied by results that are easy for the public to interpret (Kuller et al., 2021). Hydrological and hydrodynamic simulation models have proven to be indispensable for understanding physical processes in watersheds (Paiva et al., 2011, 2013; Paz et al., 2011; Pontes et al., 2015; Siqueira et al., 2018; Fleischmann et al., 2018, 2019a), and their application enables flow prediction in advance, facilitating decision-making (Collischonn et al., 2005; Fan et al., 2014; Siqueira et al., 2016, 2020).

Moreover, forecasting systems that combine two interconnected models have emerged to enhance the accuracy of predictions in complex scenarios. In this approach, a hydrological model converts forecasted rainfall into flow rates, while a hydrodynamic model represents the hydraulic behavior of a river section (Adams et al., 2018; Najafi et al., 2024). However, a common challenge in these studies lies in the practical use of forecasts, which remains underexplored. A recent study (Blöschl et al., 2019) conducted a public consultation and interviewed several researchers to identify unresolved scientific problems in hydrology. According to Blöschl et al. (2019), issues related to forecasting model uncertainties and how these uncertainties are communicated to the public and decision-makers remain significant challenges in hydrology. At local and regional levels, there is a gap in effectively communicating confidence and uncertainty in hydrological forecasts to decision-makers and the general public. The authors highlight that sister disciplines, such as meteorology, have successfully addressed these issues, exemplified by probabilistic precipitation forecasts.

In May 2024, the state of Rio Grande do Sul, in the south of Brazil, was directly affected by an unprecedented extreme rainfall resulting in mass movements and flooding events (Collischonn et al., 2025). On May 5, the unprecedented flood severely impacted the Metropolitan Region of Porto Alegre (RMPA), reaching an unprecedented level of 5.37 meters (Germano et al., 2024). This exceptional level surpassed the 1941 flood, previously considered the most impactful and traumatic in Porto Alegre's history (Possa et al., 2022). The first areas affected were the upper regions of the watershed, primarily impacting municipalities located in the Taquari and Caí river valleys, where flash floods and mass movements occurred mainly. In the RMPA, the lower part of the basin, the cities with the highest absolute numbers of directly affected people were: Canoas, 154,000; Porto Alegre, 151,000; and São Leopoldo, 84,000.

This extreme event caught communities by surprise, leaving insufficient time for evacuation in some areas. The May 2024 event could be classified as an "impossible flood" (Montanari et al., 2024). According to Montanari et al. (2024), the concept of an "impossible flood" refers to floods deemed impossible because they exceed expectations based on historical experience. Perception and preparedness for floods are shaped by past experiences and the occurrence of previous events. Therefore, identifying factors and understanding hydrological phenomena during extreme events are crucial for adapting to a future with more intense and frequent floods.

In response to this potential extreme event, a team from the Hydraulic Research Institute of the Federal University of Rio Grande do Sul (IPH/UFRGS), including the authors of this article, mobilized efforts to perform operational forecasts of Guaíba levels daily. This effort employed state-of-the-art technologies in forecasting and leveraged the best available knowledge of hydrological and hydrodynamic processes in the region available at the university. The objective was to provide advance information to aid decision-making and mitigate the flood's impacts. From May 1, 2024, forecasts were generated under various scenarios, such as a benchmark scenario with zero rainfall over the forecast horizon and scenarios considering wind effects. Thus, this study aims to present the results of the operational forecasts conducted during the extreme 2024 event, to access its accuracy and discuss lessons learned. The experience gained from this unique record event will contribute to improving future flood forecasting studies.

CASE STUDY

Study area

The Guaíba watershed covers approximately 85,000 km² and comprises five main tributaries: the Jacuí River, Taquari River, Caí River, Sinos River, and Gravataí River. It includes some of the main urban centers of Rio Grande do Sul, including the metropolitan region of Porto Alegre, the state capital. The Jacuí River is the primary tributary of the Guaíba and receives the Vacacaí-Mirim River, Upper and Lower Jacuí, and Pardo River. Additionally, near its confluence with the Guaíba, the Jacuí River merges with the Taquari River, forming a large basin. The Jacuí River flows into what is now called the Jacuí Delta, where it meets the Caí, Sinos, and Gravataí Rivers. The Guaíba then discharges into the Patos Lagoon, a lagoon approximately 250 km long and covering a surface area of about 10,000 km², extending along the state's coastal region (Lopes et al., 2018; Silveira et al., 2023). The Figure 1 illustrates the water bodies of this region.

The Guaíba basin consists of mountainous rivers with flash floods and rivers with large floodplains and slower floods. The Upper Jacuí, Taquari, and Caí river basins feature mountainous topography with narrow valleys and shallow soils. This geography results in shorter concentration times, which lead to rapid and intense floods in the basins of the Upper Jacuí, Taquari, and Caí Rivers. In contrast, the basins of the Vacacaí-Mirim, Lower Jacuí, Sinos, and Gravataí Rivers are characterized by flat regions, extensive floodplains, and numerous river meanders. These features contribute to slower water flow and longer-lasting floods.

Rainfall in the state's northern region, particularly in headwaters with shorter concentration times, rapidly reaches lower areas, increasing volumes and causing significant elevations and peaks in Guaíba levels. Rivers in the lower part of the basin (mainly the Vacacaí-Mirim and Lower Jacuí) with extensive floodplains regulate the duration of the Guaíba floods.

The Patos Lagoon also acts as a regulator of Guaíba River water levels, particularly when the wind direction opposes the natural flow. Winds blowing southward (opposite to the flow) for

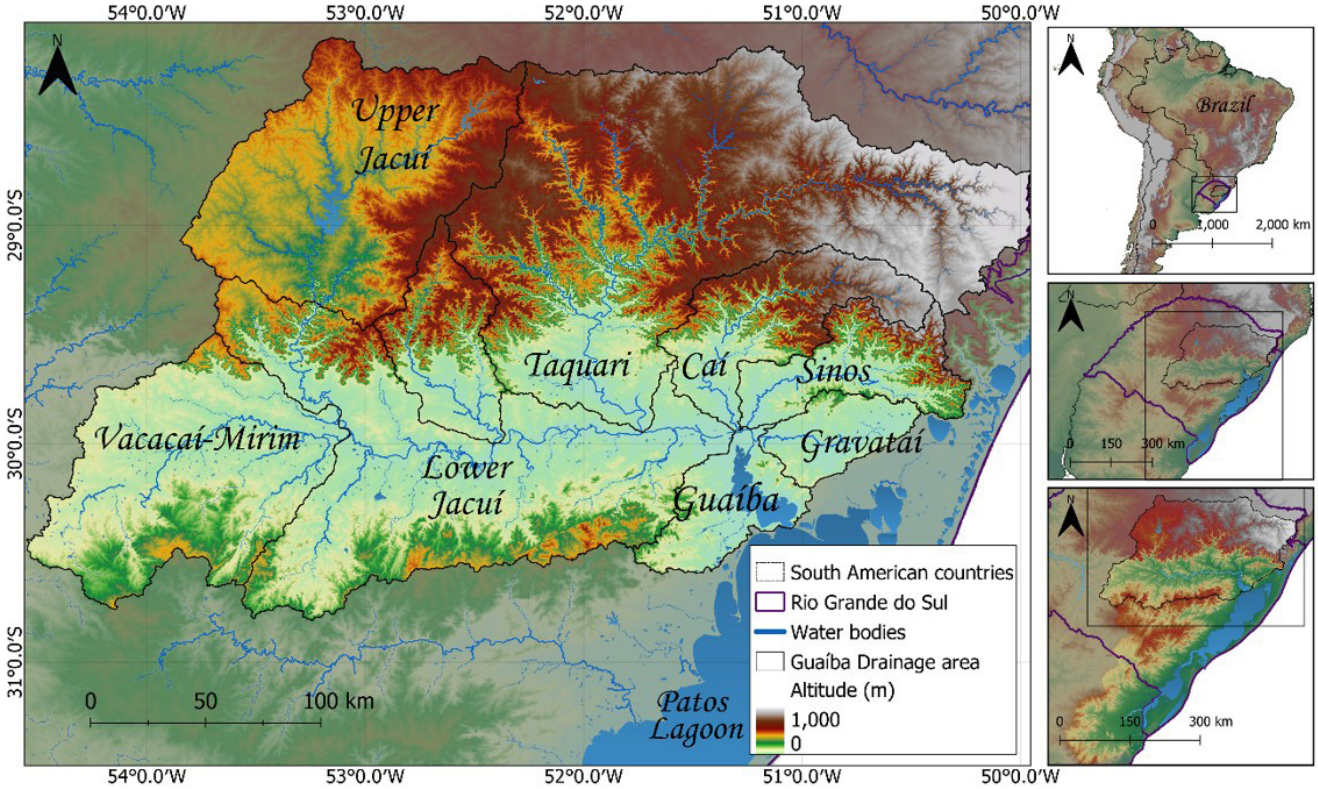


Figure 1. Location of the Guaíba River basin and its main tributaries with elevation in meters.

several hours along the Patos Lagoon can cause effect of increasing the water level in the Guaíba region near Porto Alegre, raising water levels in the Guaíba River by approximately 50 centimeters (Possa et al., 2022). With northeast winds, the opposite occurs, facilitating the dumping of the lagoon into the ocean and reducing the water levels of the Guaíba in the Porto Alegre region. The hydrodynamics of Patos Lagoon have been studied by various authors (Möller & Lorenzzentti, 1996; Moller et al., 2001; Barbosa et al., 2014; Lopes et al., 2018).

The extreme event of May 2024

Rainfall in the Guaíba River basin began mildly on April 4, leading to soil saturation in some areas. Starting on April 27, precipitation intensities increased rapidly, peaking on May 3. Collischonn et al. (2024) analyzed the rainfall during this event and concluded that it was possibly the largest registered precipitation event in Brazilian history, considering areas between 2,000 and 100,000 km² and rainfall durations ranging from 3 to 14 days. For rainfall over three days in a 10,000 km² area, the May 2024 event was 47% greater than the second-highest, demonstrating the unprecedented magnitude of the precipitation.

On May 5, the unprecedented flood reached its peak in the Guaíba River, with water levels hitting an all-time high of 5.37 meters. In the previous year, in November 2023, water levels in the Guaíba River reached 3.46 meters, exceeding the local flood threshold by only 46 centimeters (Collischonn et al., 2025). The May 2024 event far exceeded even the historic 1941 flood which reached 4.76 meters (Possa et al., 2022). Despite the metropolitan region of Porto Alegre having flood protection systems in some municipalities, in May 2024, these systems failed in several locations (Collischonn et al., 2025). Figure 2 presents a before-and-after comparison using imagery from the Copernicus Sentinel mission.

Based on a simplified analysis using observed precipitation data from the basin, measured by the National Institute of Meteorology (INMET), it was possible to identify the regions with the highest rainfall volumes. Figure 3 below shows the accumulated precipitation between April 1 and May 31, 2024. This analysis was based on raw data downloaded directly from INMET, without prior quality control or gap-filling in stations with missing values.

Even so, it is possible to observe that most of the rainfall recorded at these gauges occurred over the Upper and Lower Jacuí, Taquari, and Caí river basins. In some locations, the total accumulated rainfall during this period reached around 1000 mm.

Additionally, the figure presents observed pluviograms with an overlaid curve of cumulative precipitation over time. An average of 200 mm of antecedent rainfall can be noted prior to the May 2024 event. Another relevant aspect is that in the basins west of the Guaíba, the highest rainfall intensities and volumes occurred just before May 1. In contrast, in the northern basins, peak rainfall occurred after that date. This pattern indicates that the storm system responsible for the May disaster originated in the western region and moved northward over a few days. This type of storm led to a certain degree of synchronization in the flood peaks of the Guaíba River's tributaries.

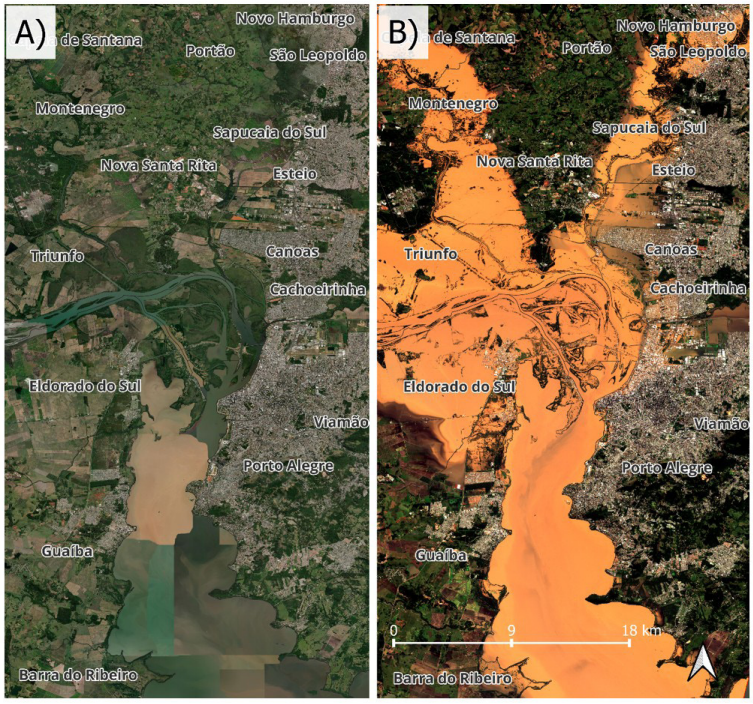


Figure 2. Copernicus images showing the before and after of the May flood: (A) Pre-flood satellite image; (B) Satellite image from May 6, 2024, shortly after the event. Souce: Copernicus Browser (2025).

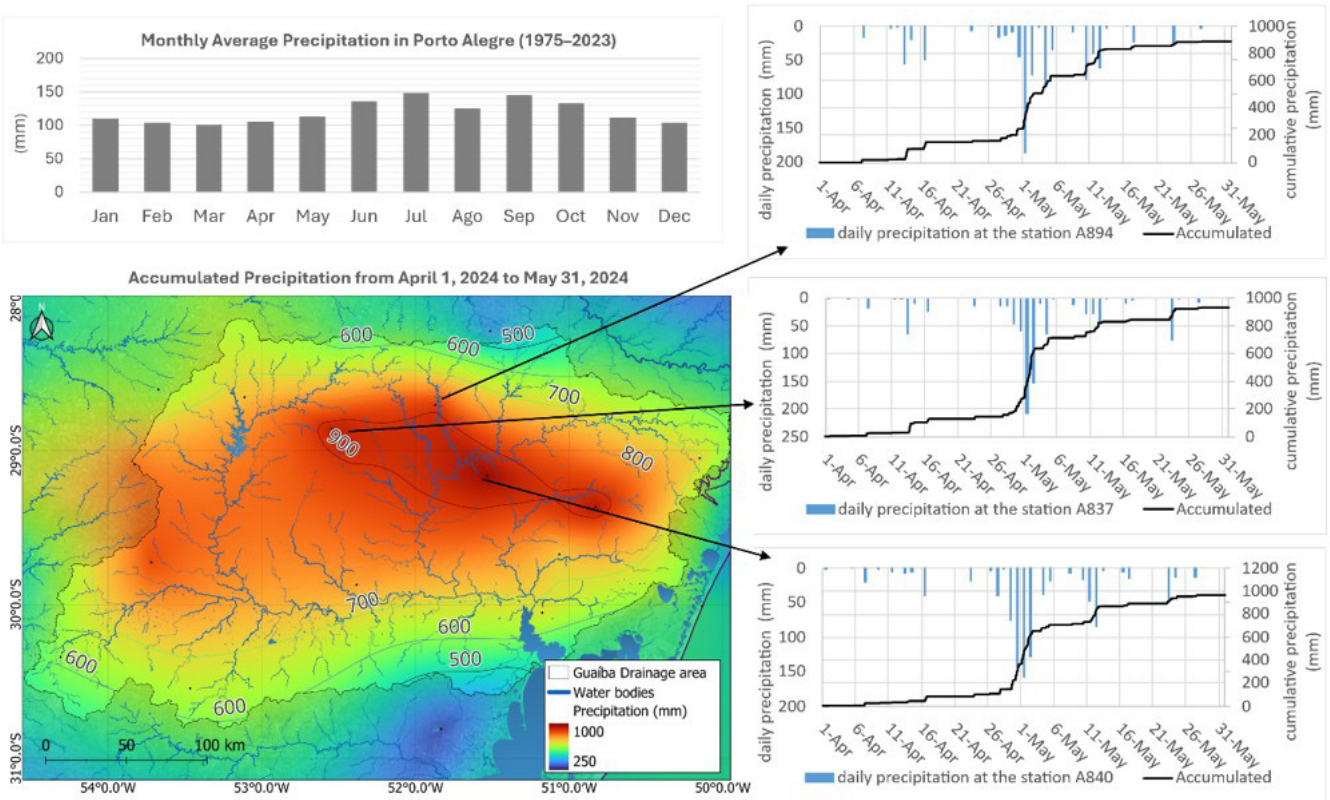


Figure 3. Precipitation analysis in the Guaíba River basin related to the 2024 flood event. The top panel shows the monthly average precipitation in Porto Alegre (1975-2023), indicating typical historical patterns. The central map presents accumulated precipitation from April 1 to May 31, 2024, based on INMET data, with totals exceeding 900 mm in parts of the Upper and Lower Jacuí, Taquari, and Caí basins. The three side panels display daily precipitation (blue bars) and cumulative precipitation (black line) for stations A894, A837, and A840.

Using hydrometric monitoring data (HidroWeb – ANA), it was possible to assess the flood magnitude at various points in the region. The heavy rainfall caused significant flooding across all rivers that feed into the Guaíba. Monitoring data showed unprecedented value throughout the event. Additionally, these data helped elucidate the hydrological dynamics of the region and the time it took for the flood wave to reach the studied points. Despite the good functionality of some stations, others failed, compromising the analysis and monitoring of the flood as it developed.

The Guaíba flood peak occurred on May 5, 2024, reaching a historic level of 5.37 meters. Discharges from the main tributaries are illustrated in Figure 4. At Point (1), the monitoring station at the Jacuí River (Station=85900000, Rio Pardo, approximately 150 km from the Guaíba), the flood peak was recorded on May 5. The attenuated and slower flood wave propagation at this location is attributed to the presence of extensive floodplains, which act to delay and dampen the peak discharge. In contrast, Points (2) and (3), monitoring stations at the Taquari River (Station=868793000, Estrela) and the Caí River (Station=87166000, Feliz), displayed different behavior due to steep valleys and shallow soils. Flood behavior in these basins showed rapid rises and recessions.

In terms of contributions, the highest discharges observed (Figure 4) are from the Lower Jacuí and Taquari rivers, contributing approximately 80% of the total discharge inflowing to the Guaíba River. Regarding peak timing, Figure 4 shows that basins with rapid runoff peaked around May 2, while basins with slower runoff peaked after May 4. Considering the Guaíba's peak occurred on May 5, the peaks were not entirely synchronized. A worse scenario could have unfolded if the peak flows had been synchronized. Nonetheless, this runoff dynamic caused rapid and substantial rises in the Guaíba's water levels.

Operational water level forecasts

As previously mentioned, operational forecasts were initiated on an emergency basis, as no forecasts existed for this region at the time. Research at IPH had already been conducted in the context of simulating Guaíba River levels, and therefore, with the imminent flood, a voluntary effort began to attempt forecasting the water levels in the Guaíba. Operational forecasts were generated daily, aiming to communicate potential scenarios for the coming days to the public. Results were presented as graphs and bulletins, which provided the public with predictions of water levels for the Guaíba.

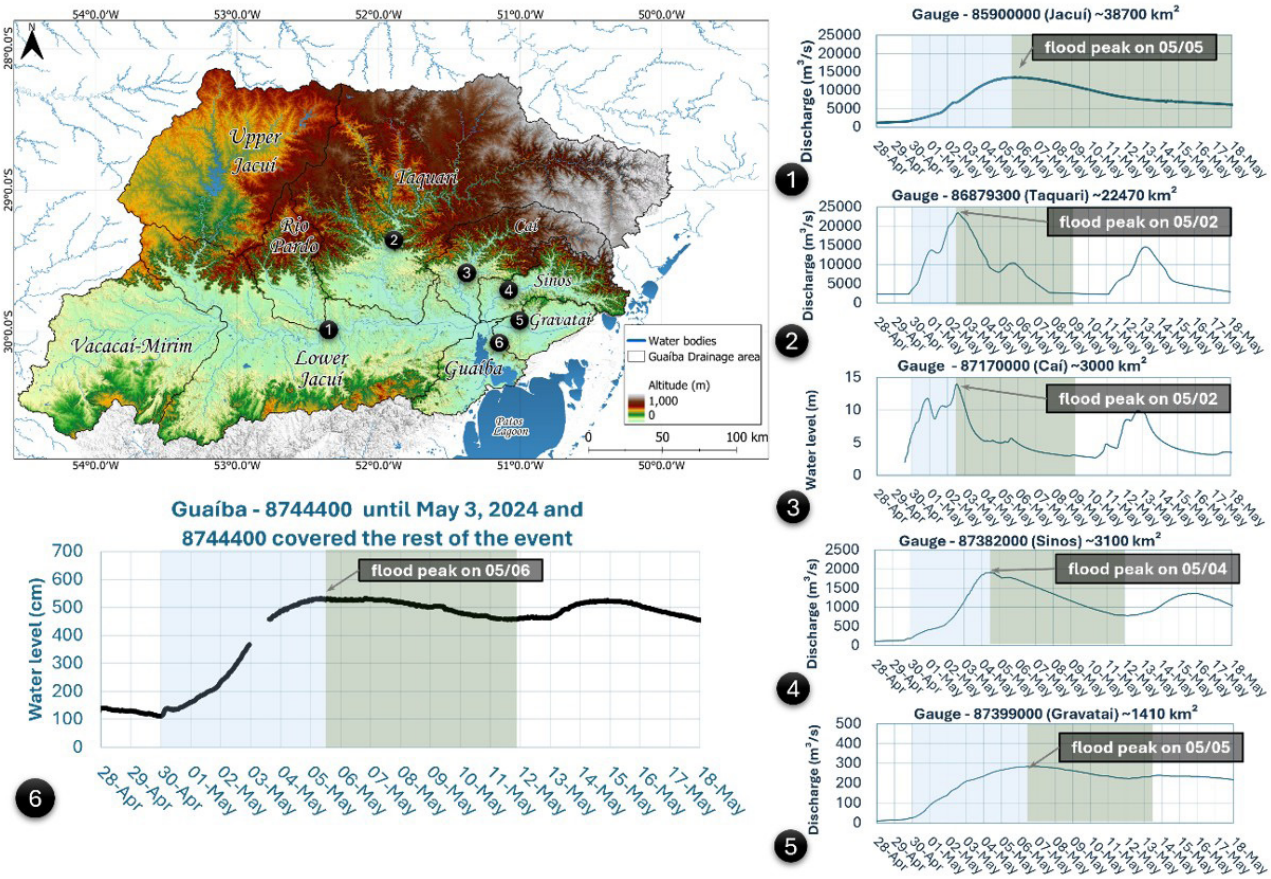


Figure 4. Observed discharges from the main tributaries of the Guaíba. Point 1: Discharges at the Rio Pardo station on the Jacuí River, with the flood peak on May 5. Points 2 and 3: Discharges and water levels on the Taquari and Caí Rivers, peaking on May 2. Point 4: Discharges from the Sinos River, peaking on May 4. Points 5 and 6: Discharges and water levels from the Gravataí River and the Guaíba, peaking on May 6 and May 5, respectively.

They were widely shared through WhatsApp, social media, and primarily on the IPH website (Universidade Federal do Rio Grande do Sul, 2025a). Additionally, television broadcasters and major newspapers also disseminated the information from the bulletins in their headlines.

Each day's forecasting process began with an analysis of the latest observations from specific hydrometric stations including river water levels and accumulated precipitation, satellite and radar imagery of recent precipitation conditions, and numerical forecasts of rainfall and wind for the next 15 days. This preliminary analysis offered an understanding of how the flood was evolving, the current state of the tributaries feeding the Guaíba, and an overview of meteorological forecasts for the next days.

The second stage involved downloading monitored hydrological data, which were input into the MGB model to simulate the hydrological behavior of the tributaries. The MGB model was employed for hydrological simulations. This model, developed by Collischonn et al. (2007), is designed to represent hydrological processes in large basins. The MGB model has been calibrated and validated for the Rio Grande do Sul Hydrological Region (RSH) (Fan et al., 2019). Discharge forecasts are generated by combining meteorological forecasts with observed precipitation and streamflow data, both historical and real-time, which feed the hydrological model to produce discharge predictions across the entire Rio Grande do Sul Hydrological Region (RSH). The MGB model accounts with a streamflow data assimilation approach first implemented by Rolim et al. (2007) and later improved by Meller et al. (2012) and Fan et al. (2015). However, the data assimilation approach was not used during the forecast's elaboration since several gauging stations that could be used for data assimilation was damaged or not properly operating due to the unprecedented rainfalls and floods. Through the flood forecasts preparation this was not exactly a major limitation, given that the model has relatively good accuracy on flow representations.

The resulting hydrological model discharges were input into the hydrodynamic model. Hydrodynamic simulations were performed using the unsteady 1D flow simulation component of the HEC-RAS model (U.S. Army Corps of Engineers, 2021). The digital elevation model used was generated by combining aerial imagery from the Rio Grande do Sul State Spatial Data Infrastructure (IEDE-RS) database, provided by the State of Rio Grande do Sul. These data are owned by the state and were used with the necessary authorizations for academic purposes. Bathymetric data were used to represent the channel beds, previously provided by technical reports and publicly available documents on the internet for the Sinos, Caí, and Jacuí rivers. For the Guaíba River and Lagoa dos Patos, bathymetric data based on the Brazilian Navy's hydrographic and navigation charts, derived from surveys conducted between 1961 and 1966, were used, as referenced in Lopes et al. (2018). Others areas were supplemented with the SRTM digital elevation model, which has a spatial resolution of approximately 90 meters (Farr et al., 2007).

The simulation domain encompasses the upstream portion of the Guaíba, beginning in the Jacuí River, passing through the Patos Lagoon, and extending to its outlet in Rio Grande, where the downstream boundary condition is applied. Lateral inflows to the 1D model include the discharges from the Taquari, Caí, Sinos, and Gravataí rivers, along with wind data to represent the effects of water retention caused by southern winds.

Although the Guaíba and Patos Lagoon system includes large lacustrine and floodplain areas, the behavior of the system during the May 2024 flood event was predominantly fluvial. The flood wave propagated as a riverine flow, characterized by high velocities and well-defined flow direction. Recent field measurements reported by Andrade et al. (2024) near downtown Porto Alegre recorded streamflow velocities on the order of 4 m/s, reinforcing the fluvial nature of the event.

Given this behavior, the 1D hydrodynamic model was considered appropriate for representing the main flow dynamics along the river-lake system. In addition, the downstream boundary condition for the model was defined at a point near the outlet to the Atlantic Ocean, sufficiently distant from the region of interest (the Guaíba River), within the complex Patos Lagoon system. This approach allowed the potential simplification errors in the lagoon region to dissipate before influencing the simulation results in the upstream Guaíba reach. Therefore, despite inherent limitations, the 1D approach proved adequate for operational forecasting of water levels during the event.

The 1D hydrodynamic model was not fully calibrated prior to the event. Instead, it was progressively calibrated as the flood unfolded, in response to the urgent operational needs. Initial configurations were based on preliminary tests and visual assessments, which provided a satisfactory starting point. As the event evolved, the model was incrementally improved, including the addition of cross sections and adjustments to Manning's roughness coefficients. Final calibrated values were 0.0375 for the main channel and 0.6 for the floodplain areas. Despite the lack of a formal pre-event calibration, these refinements allowed the model to achieve reliable performance during the real-time forecasting period.

Post-processing techniques were limited to the application of simple offset corrections to reduce systematic bias between simulated and observed water levels. These adjustments ensured that each forecast started from a level consistent with the most recent observed data, without altering the dynamic response of the model.

Prior to 2023, there was no forecasting system capable of predicting water levels for the Guaíba River. To address this gap, research began at the Hydraulic Research Institute (IPH) at the Federal University of Rio Grande do Sul, focusing on integrating hydrological and hydrodynamic models with meteorological forecasts for this area. The primary objective was to study the phenomena associated with the Guaíba River system and develop tools to improve forecasting capabilities in a region that previously lacked such resources and had already experienced significant flooding in the past.

This effort culminated in the creation of a framework designed to provide accurate predictions, supporting flood mitigation and preparedness. Each water level forecast's results were translated into clear and actionable recommendations to assist decision-making by local authorities and stakeholders. However, the unprecedented nature of the May 2024 flood event presented significant challenges, requiring local expertise and a deep understanding of the system's dynamics as the event unfolded.

Scenarios were designed considering different approaches, including benchmark scenarios, such as assuming zero rainfall and wind within the forecast horizon, representing an optimistic outlook, as well as conservative scenarios that accounted for extreme effects on the system, such as prolonged intense winds over the Patos Lagoon. Additionally, scenarios with varying precipitation forecasts were also analyzed to assess the spatial and temporal agreement of meteorological predictions and the potential conditions for the coming days. Teamwork was crucial for analyzing scenarios and addressing these challenges. Figure 5 presents a simplified schematic of the forecasting process. The diagram illustrates the forecasting process. It began with the download and analysis of observed data and upcoming meteorological forecasts, followed by hydrological and hydrodynamic modeling. Finally, the results were analyzed, and informational bulletins were written to interpret these forecasts. All messages were recommendations based on the forecasters' interpretations. Official statements on behalf of Civil Defense were not issued. The bulletins aimed solely to inform what could happen in the coming days regarding water levels in the Guaíba.

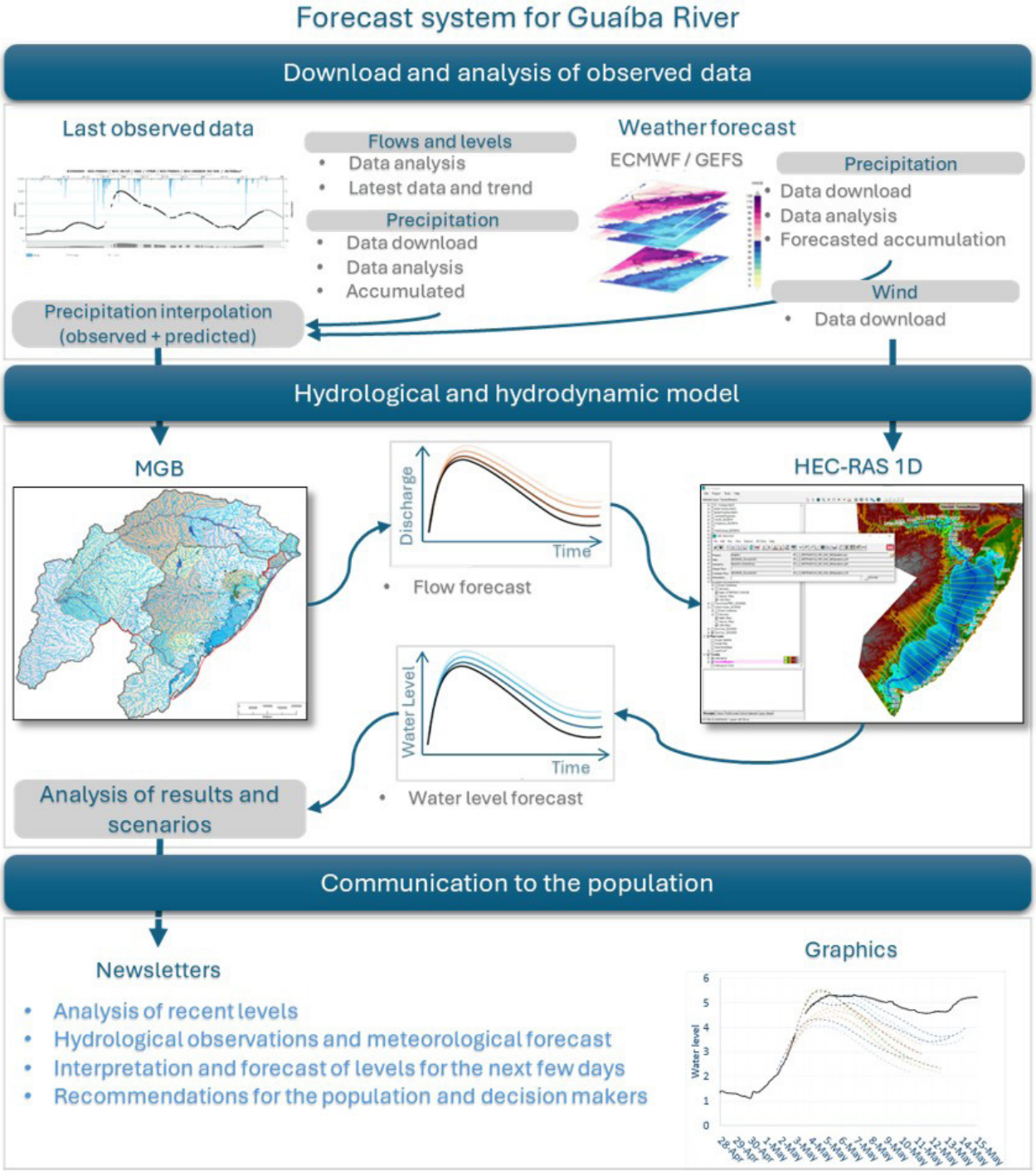


Figure 5. Simplified schematic of the forecasting process.

Meteorological forecasts used as input for the hydrological and hydrodynamic models were obtained from two global numerical weather prediction systems: the European Centre for Medium-Range Weather Forecasts (ECMWF) and the Global Ensemble Forecast System (GEFS). These forecasts were used without post-processing, as they were accessed in real time during the event. The spatial resolution was approximately 0.1° for ECMWF and 0.25° for GEFS, with temporal resolution of 6 hours and a forecast horizon of up to 10 days.

Wind data were considered both from observed records and forecasts. Observed wind conditions were obtained from INMET (Brazilian National Institute of Meteorology) and local automatic weather stations near the Patos Lagoon. Forecasted wind fields were also taken from ECMWF and GEFS, used primarily to simulate potential backwater effects in the Guaíba River due to persistent southerly winds. These data were directly used in the hydrodynamic model without additional filtering, given the operational context in which forecasts were being generated.

Although the forecasting system used in this study is robust and technically validated, the forecasts were conducted under emergency conditions, given the unprecedented nature of the May 2024 flood event. Throughout the forecasting period, several rainfall and streamflow stations were used, but some had to be discarded due to operational failures. Data were obtained and validated from various sources, including the Brazilian Geological Service (SGB), which provided critical field information during the flood.

On the first forecasting day, the main message was that this flood would surpass previous flood events, requiring the activation of the city's flood protection system, including the closure of floodgates as part of the city's flood defense infrastructure. On the second day, forecasts already suggested a flood exceeding the 1941 event (referencing the 4.76-meter level). The third day was concerning that scenarios confirmed predictions of a 5-meter flood level in the Guaíba. On the fourth and fifth days, the forecasts were again confirmed, with the Guaíba reaching 5.37 meters (Collischonn et al., 2025; Germano et al., 2024). One additional information relevant at these cases was that the peak values were below 6 meters, which was the theoretical top level of the Porto Alegre capital city protection system. Moreover, the flood proved to be long-lasting based on interpretations of tributary data. On the sixth day, the first warning of a possible secondary peak in the Guaíba was issued (Figure 6). Forecast scenarios indicated a second rise in eight days in advance. In Figure 6, two forecasts are presented: one issued on May 6, 2024, and another on May 8, 2024. In both panels, the second water level peak is already visible in the forecast. Only two forecast scenarios were communicated to the public, as the remaining scenarios indicated similar conditions.

Figure 7 shows four bulletin graphics issued during the flooding event. In addition to graphs, textual explanations were included to assist in interpreting the forecasts. The emergency operational forecasting framework not only provided valuable information for decision-making (GaúchaZH, 2024a), but also underscored the importance of clear and effective communication with the public. The forecasts were issued daily with a time horizon of up to 15 days. From Authors knowledge various organizations relied on these forecasts to plan actions during the event. All forecasts were published and are accessible at Universidade Federal do Rio Grande do Sul (2025a, 2025b).

From the seventh to the twelfth days, forecasts confirmed a reduction in level and the secondary peak, indicating renewed flooding in already affected areas. On the thirteenth day, all scenarios pointed to rising levels near 5 meters again, was confirmed the following day.. On May 14, the secondary peak of 5.20 meters was observed, with forecasts indicating stability at high levels. From the fifteenth day onward, forecasts aligned with a slow recession in levels, with occasional small rises caused by local rainfall and southern wind effects. These episodes occurred and were forecast on May 24, May 28, and June 4, 10, and 15. The Figure 8 presents a representation of the effects mentioned and predicted over the course of the forecasts based on the observed data. Additionally, in Figure 8, it is possible to observe the different periods of rapid water level rises (caused by local rainfall and effects generated by the south wind), as well as the extended periods of slow recession of the flood.

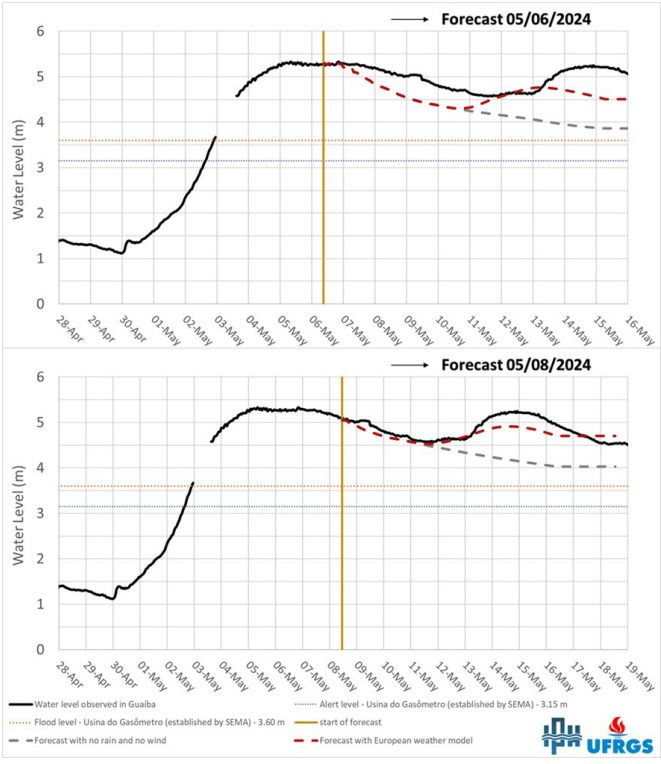


Figure 6. Forecasted and observed water levels in the Guaíba River. Top: forecast issued on May 6, 2024. Bottom: forecast issued on May 8, 2024. The black line shows the observed water level in the Guaíba River. The red dashed line represents the forecast using the European weather model, while the gray dashed line shows the scenario assuming no rainfall and no wind. The vertical yellow line indicates the forecast start date. The flood and alert thresholds, established by SEMA at the Usina do Gasômetro gauge station, are marked with red and blue dotted lines, respectively.

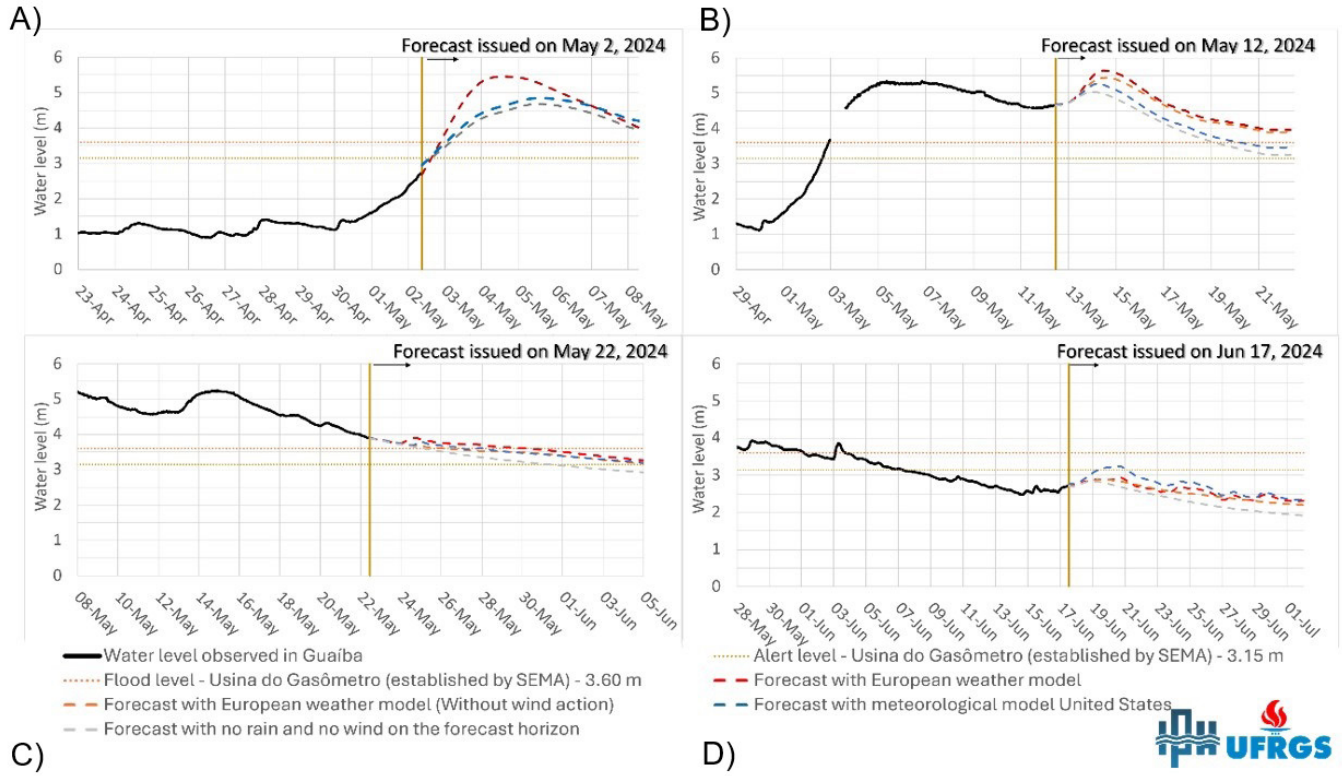


Figure 7. Guaíba water level forecasts issued on different days during the event. (A) Forecast from May 2, indicating levels above 5 meters; (B) Forecast from May 12, showing a secondary peak on May 15; (C) and (D) Forecasts from May 22 and June 17, indicating a slow recession in Guaíba levels.

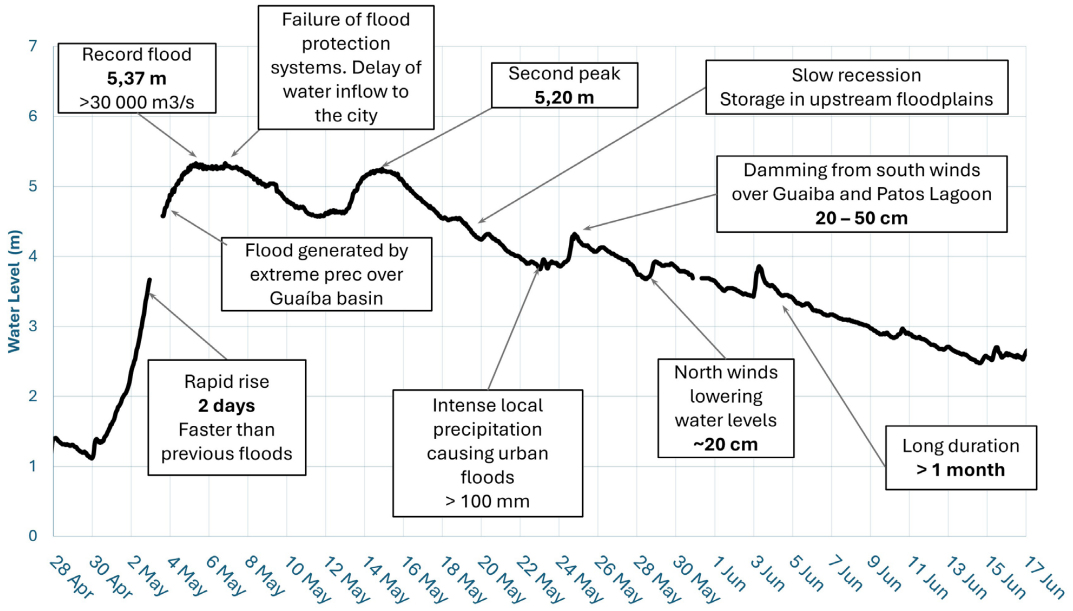


Figure 8. Main events observed throughout the forecasts based on the data observed in Guaíba.

METHODS FOR EVALUATING THE FORECASTS

This study evaluated the forecasts using two metrics: the Nash-Sutcliffe Efficiency coefficient (NSE) and the Root Mean Square Error (RMSE). The methods are described in the subsections below. The forecast began on May 1, 2024, and continued until July 11, 2024. Metrics, however, were calculated only for the period from May 1 to June 24, as this corresponds to the active phase of the flood, during which water levels were above the flood threshold. After June 24, water levels fell below the flood threshold, and forecasts focused on monitoring gradual recession.

Additionally, the forecasting bulletins were individually evaluated for the same period, considering the text messages delivered to the public. This analysis sought a novel approach to assessing forecast accuracy by considering not only the numerical values produced by the models but also the interpretation and communication by the forecasters.

Nash-Sutcliffe Efficiency coefficient (NSE)

The Nash-Sutcliffe Efficiency (NSE) coefficient relates the error of the evaluated forecast to the error of a forecast using the long-term mean of the observed data. The value is dimensionless and ranges from -∞ to 1. Values above 0.5 are considered satisfactory, those above 0.65 are deemed good, and values above 0.75 are considered very good (Fleischmann et al., 2019b). Values near zero indicate that the model performs as poorly as the mean. Positive values suggest that the model outperforms the mean, while negative values indicate it performs worse. The NSE efficiency coefficient is calculated using the following equation:

$$NSE_h = 1 - \frac{\sum_{n=1}^{N} (Q_{Phn} - Q_{Ohn})^2}{\sum_{n=1}^{N} (Q_{Ohn} - O_m)^2} \quad (1)$$

where: N is the number of time intervals; n is a given forecast being evaluated; h is the forecast horizon being evaluated; QPhn is the forecasted value at time interval n for horizon h; QOhn is the observed value for the same interval and horizon; and Om is the mean of observed values over N time intervals.

While the traditional Nash-Sutcliffe Efficiency (NSE) remains one of the most widely used performance metrics for hydrological modeling, it presents limitations in forecasting contexts. This is because it evaluates model performance relative to the mean discharge. In short-term forecasting applications, a more appropriate reference could be a persistence model, which assumes that discharge remains constant from the time of forecast issuance. Although not applied in this study, the use of persistence-based performance metrics is encouraged for future improvements.

Root Mean Square Error (RMSE)

The Root Mean Square Error (RMSE) measures the average difference between observed values and corresponding forecasts. The mathematical formulation of the RMSE is given by the equation:

$$RMSE_h = \sqrt{\frac{1}{N} \sum_{n=1}^{N} (Q_{Ph,n} - Q_{Oh,n})^2} \quad (2)$$

This coefficient represents the magnitude of the error in the simulation, giving greater weight to extreme values. Better results are indicated by values closer to zero.

Evaluation of bulletins and forecast communication

To assess the accuracy of the bulletins, each bulletin was individually analyzed for its key messages and recommendations related to the forecasts. These messages were categorized daily throughout the forecasting period to identify whether they occurred as predicted. A message was classified as accurate if the forecast stated in the text materialized within the forecast horizon.

For example, the message "Forecast scenarios indicate a reduction in Guaíba levels below the alert threshold by the weekend" would only be classified as accurate if, during the subsequent weekend, the water level indeed dropped below the alert threshold. Otherwise, it would be classified as inaccurate. The evaluation of the bulletin is detailed in the discussion chapter regarding forecasting accuracy.

RESULTS

From May 1, 2024, forecasts for Guaíba River water levels were generated. These ensemble forecasts considered various scenarios, such as with and without rainfall in the forecast horizon, as well as scenarios accounting for wind effects. The forecasts are shown in Figure 9.

When comparing all forecasts to the observed data, most critical scenarios successfully reproduced the general behavior of the flood event, particularly the timing and magnitude of peak and recession phases. This alignment indicates that the forecasting system was able to capture the dominant hydrological responses during the event. Additionally, scenarios assuming no rainfall or wind in the forecast horizon consistently produced water levels lower than those observed. This outcome is explained by the physical role that rainfall and wind play in the flood dynamics of the region. In the initial hours of the forecast horizon, the influence of new rainfall or wind events is indeed limited. However, beyond the first 24 hours, these forcings become increasingly relevant. Precipitation in upstream tributaries leads to additional inflows that propagate toward the Guaíba with a certain time lag, while strong southerly winds over the Patos Lagoon can cause a backwater effect, elevating water levels in the Guaíba.

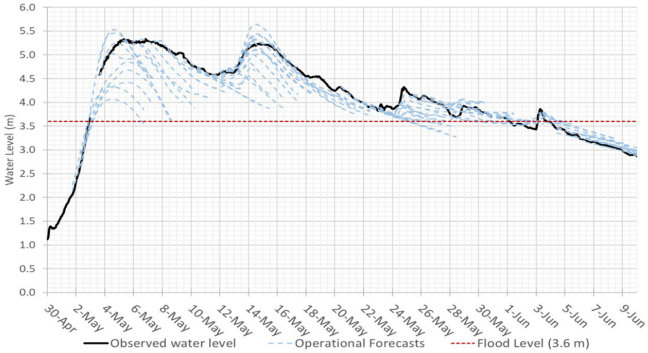


Figure 9. Operational forecasts (blue dashed lines) plotted alongside observed data (solid black line) and the flood threshold of 3.6 meters (red dashed line).

Therefore, forecasts that exclude rainfall and wind tend to underestimate water levels over longer horizons, as they disregard these key hydrometeorological drivers. These simplified scenarios are useful as lower-bound references but are not physically realistic representations of expected conditions during an active flood event.

Figure 10 shows the NSE and RMSE results for each forecast type. In both metrics, the results showed good accuracy within the first 24 hours of the forecast horizon. Beyond 24 hours, a decline in NSE values was observed for the "zero rainfall and zero wind" forecasts, indicating a dependency on these variables for predicting Guaíba levels at these horizons. This reduction, after approximately one day, can be attributed to the propagation time and distance of tributaries to the Guaíba. Forecasts that incorporated meteorological models showed optimal accuracy (NSE above 0.7 across nearly the entire forecast horizon), demonstrating better anticipation of events overall. The ensemble mean also performed well, closely matching meteorological model results.

RMSE, calculated for each forecast horizon, varied similarly. Errors ranged from 10 to 70 centimeters. Forecasts assuming zero rainfall and wind showed errors exceeding 50 cm for horizons beyond four days (96 hours). In contrast, scenarios using meteorological models showed average errors of about 10 cm within the first 24 hours, indicating excellent short-term accuracy.

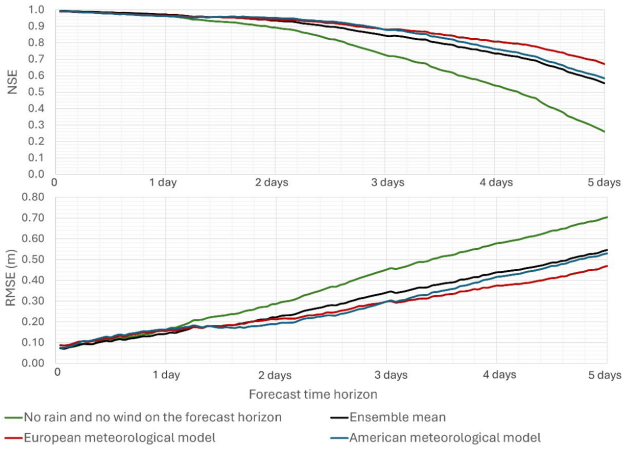


Figure 10. NSE and RMSE performance metrics for each forecast horizon under different scenarios: Zero rainfall and wind (green line); European Model (red line); American Model (blue line); and Ensemble Mean (black line).

Discussion on forecast accuracy and value

Forecast bulletins were distributed to the community as text accompanied by forecast figures. These messages aimed to provide the public with an understanding of the simulated scenarios. They also recommended actions for both potentially affected populations and decision-makers. To illustrate these messages, a series of reports published by a local communication channel that used the forecast bulletin for their headlines were selected. Figure 11 presents a summary of some key messages published by this local communication channel. During the event, critical aspects of the flood were predicted, allowing timely public warnings.

To analyze the communications issued during the forecasts, a summary of the main statements from the bulletin was compiled and compared with actual events. Additionally, alongside each statement, comments and the lead time of the predicted event were included. Alerts issued during the event are summarized in Table 1.

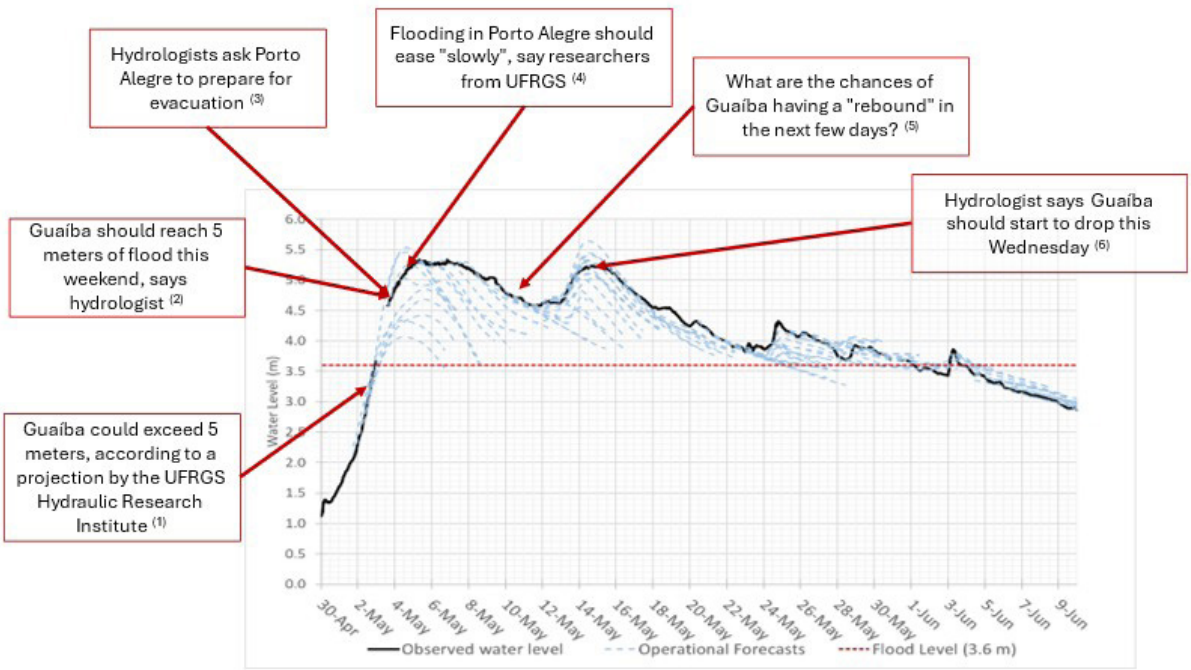


Figure 11. Key headlines based on forecasts published by a regional press outlet. References: (1) GaúchaZH (2024a); (2) GaúchaZH (2024b); (3) GaúchaZH (2024c); (4) GaúchaZH (2024d); (5) GaúchaZH (2024e); (6) GaúchaZH (2024f).

Table 1. Main alert phrases issued by the operational flood forecasting system during the 2024 flood event in the Guaíba River region, which were disseminated to the public and authorities throughout the forecasting period.

Date	Main Message	Occurred	Lead Time	Comment
May 1	Could exceed the November 2023 event and approach the 1941 flood level	Yes	~72h (3 days)	Surpassed the 1941 flood level on May 3
May 2	Many scenarios suggest levels equal to or higher than the 1941 mark	Yes	24h	Surpassed the 1941 flood level on May 3
May 3	Recommendation for evacuation preparedness measures in regions potentially affected in Porto Alegre	Yes	-	Evacuation recommendations were well followed
May 5	Forecasts indicate a prolonged flood event	Yes	Entire horizon	
May 6	Could reach 5 m if rainfall forecasts in the Guaíba river basin are confirmed	Yes	~192h (8 days)	First warning of a potential resurgence for the week of May 14
May 7	Possibility of resurgence due to forecasted rainfall starting at the weekend, potentially reaching the 5 m mark	Yes	~168h (7 days)	Second warning about the resurgence for May 14 (1 week prior)
May 8	Elevated levels above 5 m expected in the coming days	Yes	~144h (6 days)	Third warning about the resurgence for May 14 (1 week prior)
May 9	Possibility of resurgence due to forecasted rainfall starting at the weekend, potentially reaching the 5 m mark	Yes	~120h (5 days)	Fourth warning about the resurgence for May 14 (1 week prior)
May 12	Confirmation of the flood resurgence with new levels above 5 m	Yes	~48h (2 days)	Warning about the resurgence for May 14
May 13	Maximum level expected between Monday and Tuesday	Yes	~24h	Reached 5.23 m on Tuesday, May 14
May 14	Forecast scenarios indicate stabilization at levels above 5 m after the resurgence	Yes	~48h (2 days)	
May 14	Slow recession expected over the next few days, remaining above 4 m throughout the week	Yes	~168h (7 days)	Levels stayed above 4 m for at least 1 week
May 16	Slow reduction of levels below 5 m	Yes	~48h (2 days)	
May 16	Expected to stay above 4 m until the start of next week	Yes	~120h (5 days)	
May 16	Remaining above the flood level of 3 m until the end of the month or beyond	Yes	Entire horizon	
May 20	Possible Guaíba river impoundment due to strong south winds expected on Friday, causing a new rise to around 4 m	Yes	~96h (4 days)	Warning about impoundment due to wind effects, raising the Guaíba to 4.3 m on May 24
May 23	Rainfall during the week could also contribute to an increase in levels, prolonging the flood above 3 m into June	Yes	Entire horizon	
May 24	Guaíba river levels expected to rise due to forecasted winds, maintaining high levels	Yes	-	Confirmed impoundment due to wind effects, raising the Guaíba to 4.3 m on May 24
May 28	The Guaíba river level is expected to reach the flood level defined for the Usina do Gasômetro station in the next few days	Yes	~72h (3 days)	3 days later, the flood level was reached
May 29	The Guaíba river level is expected to reach the flood level defined for the Usina do Gasômetro station in the next few days	Yes	~48h (2 days)	2 days later, the flood level was reached
May 31	The Guaíba river level is expected to reach the flood level defined for the Usina do Gasômetro station in the next few days	Yes	~24h	1 day later, the flood level was reached on June 1

Table 1. Continued...

Date	Main Message	Occurred	Lead Time	Comment
June 3	The Guaíba river is expected to reach the flood level defined for the Usina do Gasômetro station starting Tuesday	Yes	-	The level was reached on June 3 at 5:00 PM, confirming the forecast
June 3	May approach the alert level by the end of the week	Yes	~96h (4 days)	On June 7 (Saturday), the alert level of 3.15 m was reached
June 4	The Guaíba river is expected to reach the alert level defined for the Usina do Gasômetro station by the end of the week	Yes	~72h (3 days)	On June 7 (Saturday), the alert level of 3.15 m was reached
June 5	The Guaíba river is expected to fall below the alert level defined for the Usina do Gasômetro station in the next few days	Yes	~48h (2 days)	On June 7 (Saturday), the alert level of 3.15 m was reached
June 6	The Guaíba river is expected to fall below the alert level defined for the Usina do Gasômetro station between today and tomorrow	Yes	~24h	On June 7 (Saturday), the alert level of 3.15 m was reached
June 6	It is expected to remain below the alert level for the next 10 days	Yes	Entire horizon	Guaíba returned to the alert level in the early hours of June 19-20
June 12	South winds from the weekend cause impoundment	Yes	~48h (2 days)	Impoundment action and slight increase in levels observed
June 14	Forecast scenarios indicate stabilization/increase in Guaíba river levels	Yes	Entire horizon	Increase caused by south wind effects and precipitation volumes
June 14	Increase in Guaíba river levels expected throughout the week	Yes	~120h (5 days)	Increase caused by south wind effects and precipitation volumes
June 14	Forecasts do not indicate levels above the alert level	No	-	Levels surpassed the alert level on June 19
June 16	New forecast scenarios indicate a possible increase in Guaíba river levels throughout the week	Yes	~96h (4 days)	Increase caused by south wind effects and precipitation volumes
June 16	Could exceed the alert level in the worst-case scenario	Yes	~96h (4 days)	Levels surpassed the alert level on June 19
June 17	Forecast scenarios indicate an increase in Guaíba river levels in the coming days	Yes	~48h (2 days)	Increase caused by south wind effects and precipitation volumes
June 17	Could exceed the alert level in the worst-case scenario	Yes	~48h (2 days)	Levels surpassed the alert level on June 19
June 18	Expected to reach values around the alert level	Yes	~24h	Levels surpassed the alert level on June 19
June 19	Forecast scenarios indicate maintenance or further increase in Guaíba river levels in the coming days	Yes	~12h	
June 19	Expected to remain around or above the alert level	Yes	~12h	Guaíba returned to the alert level in the early hours of June 19-20
June 20	Followed by a reduction below the alert level by the weekend	No	-	Levels slightly exceeded the alert level over the weekend of June 22-23, reaching 3.22 m
June 21	Forecast scenarios indicate a reduction in Guaíba river levels below the alert level by the weekend	No	-	Levels slightly exceeded the alert level over the weekend of June 22-23, reaching 3.22 m
June 21	South wind event between Sunday and Tuesday, causing another rise	Yes	~72h (3 days)	Impoundment action observed

Upon examining Table 1, some events were forecast with up to 72 hours' lead time, demonstrating excellent quantitative accuracy. For instance, on May 1, the following statement was published:

> The flood threshold of 3 meters is expected to be surpassed by Thursday afternoon (05/02/2024), leaving little time to close the flood protection gates. Additionally, water levels are expected to rise further, potentially exceeding 4 meters between Friday and Saturday. If this occurs, it may surpass the November 2023 event and approach the 1941 flood level (Medeiros, 2025).

With approximately 72 hours' lead time, it was possible to predict levels surpassing the last historic flood in 1941. For subsequent rises, such as the secondary peak on May 14, the May 6 bulletin stated:

> Forecast scenarios indicate a prolonged flood. Elevated levels above 5 meters are expected in the coming days. A slow decline will maintain levels above 4 meters throughout the week. Rainfall expected over the weekend could raise levels back to 5 meters (Medeiros, 2025).

This forecast, with about 192 hours (8 days) of lead time, was crucial as many residents began returning to their homes after a gradual water reduction, unaware of the potential secondary peak. The risk warning helped the population remain alert.

In general, no false alarms were identified throughout the event. All high-risk and impactful messages that forecast the occurrence of critical events were confirmed. However, after the slow recession of floodwaters and due to uncertainties related to wind effects, some messages showed lower accuracy—mainly due to the non-detection of subsequent rises in water levels. Nevertheless, since the flood had already lost intensity during this phase, these undetected events did not cause significant impacts or undermine the credibility of the forecasts.

Finally, it is important to highlight that this accuracy was achieved through the interpretation of the system by the forecasters. Signals of rising or falling water levels, in time horizons greater than 72 hours, were interpreted and discussed among the forecasters. Other variables were used to support the description of phenomena in the bulletin, such as the agreement (or lack thereof) of meteorological models and the current conditions of water levels in rivers. During the event, several difficulties arose, such as failures in real-time monitoring and uncertainties in the models. This situation was alleviated through intense discussions and interpretations carried out by the forecasters in addition to significant adjustments and recalibrations of the models were carried out throughout the process. This experience underscored the importance of investing in people, as technologies, no matter how advanced, cannot replace the critical analysis and contextualization done by experts

CONCLUSIONS

This study aimed to present the results of the operational forecasts conducted during the extreme flood event of May 2024 in southern Brazil, evaluating the accuracy of the hydrological and hydrodynamic forecasting system, and discussing the lessons learned. By combining hydrological and hydrodynamic models with meteorological forecasts, we were able to accurately predict critical water levels in the Guaíba River, including flood scenarios with an operational lead time of three days. The results indicate that the model was able to represent the system and anticipate the impacts of extreme floods.

The unprecedented nature of the event demanded a paradigm shift, emphasizing the need for local knowledge and interpretation of system physics as the event unfolded. This study highlights the importance of integrated hydrological forecasting approaches to anticipate and mitigate floods.

The results demonstrate that forecasts incorporating precipitation and wind data achieved better performance, highlighting the physical role of these variables within the system. Furthermore, performance metrics such as NSE and RMSE showed good results for forecasts with a 5-day horizon. Longer forecast horizons would likely exhibit lower performance, indicating a dependency on the accuracy of meteorological prediction models.

These results, coupled with the clarity and precision of the forecasts, translated into efficient predictions with significant lead times for critical events, enabling better preparedness and response from authorities and communities.

The study highlights the need for continued improvement of forecasting models and investment in monitoring infrastructure to address future extreme events. Challenges such as real-time observation failures and model uncertainties underscore the importance of involving local experts and technicians in interpreting forecasts. Lessons learned from recent events suggest that effective flood management requires an integrated approach, including collaboration among scientists, authorities, and the community. This approach can enhance resilience in affected regions and mitigate the impacts of future floods.

Finally, it is important to highlight that these satisfactory results and the positive impact of the forecasts were achieved even in the absence of a formal, structured flood alert system. There was no pre-established institutional framework for flood forecasting and communication in the region, nor prior community preparedness, training, or standardized protocols for message dissemination. Despite these limitations, the forecasts proved effective in supporting emergency decision-making and informing the population during a critical event. This reinforces the potential of operational hydrometeorological forecasting systems, even under improvised conditions, and underscores the urgency of institutionalizing and strengthening such systems for future events.

ACKNOWLEDGEMENTS

We extend our sincere gratitude to the Geological Survey of Brazil (SGB) and the Brazilian National Water and Sanitation Agency (ANA) for providing hydrological data through operational systems and personal communications. We also thank the National Electric System Operator (ONS) for granting access to meteorological forecasts via the SINtegre platform. Our appreciation goes to the Hydraulic Research Institute (IPH) for infrastructure support and to our colleagues at IPH for their valuable assistance throughout the forecasting efforts. The author F. M. F. thanks the Brazilian CNPq (Conselho Nacional de Desenvolvimento Científico e Tecnológico) for supporting this research under Grant Number 304973/2022-0.

DATA AVAILABILITY STATEMENT

Research data is only available upon request.

REFERENCES

Adams III, T. E., Chen, S., & Dymond, R. (2018). Results from operational hydrologic forecasts using the NOAA/NWS OHRFC Ohio River Community HEC-RAS Model. *Journal of Hydrologic Engineering*, *23*(7), 1. http://doi.org/10.1061/(ASCE)HE.1943-5584.0001663.

Alfieri, L., Burek, P., Dutra, E., Krzeminski, B., Muraro, D., Thielen, J., & Pappenberger, F. (2013). GloFAS-global ensemble streamflow forecasting and flood early warning. *Hydrology and Earth System Sciences*, *17*(3), 1161-1175. http://doi.org/10.5194/hess-17-1161-2013.

Andrade, M. M., Piazera, M., Luz, R., Nunes, J. C. R., Scottá, F., & Silva, T. (2024). Flow measurements with ADCP on the Guaíba River, during the highest water level recorded in history-May 2024 (floods in the State of Rio Grande do Sul, Brazil). *Revista Brasileira de Recursos Hídricos*, *29*, e36. http://doi.org/10.1590/2318-0331.292420240052.

Barbosa, R., Cavalcante, L., André, C., & Mendes, B. (2014). Calibração e validação do módulo de correntologia do modelo IPH-A para a Laguna dos Patos (RS/Brasil). *Revista Brasileira de Recursos Hídricos*, *19*(3), 191-204. http://doi.org/10.21168/rbrh.v19n3.p191-204.

Blöschl, G., Bierkens, M. F. P., Chambel, A., Cudennec, C., Destouni, G., Fiori, A., Kirchner, J. W., McDonnell, J. J., Savenije, H. H. G., Sivapalan, M., Stumpp, C., Toth, E., Volpi, E., Carr, G., Lupton, C., Salinas, J., Széles, B., Viglione, A., Aksoy, H., Allen, S. T., Amin, A., Andréassian, V., Arheimer, B., Aryal, S. K., Baker, V., Bardsley, E., Barendrecht, M. H., Bartosova, A., Batelaan, O., Berghuijs, W. R., Beven, K., Blume, T., Bogaard, T., Borges de Amorim, P., Böttcher, M. E., Boulet, G., Breinl, K., Brilly, M., Brocca, L., Buytaert, W., Castellarin, A., Castelletti, A., Chen, X., Chen, Y., Chen, Y., Chifflard, P., Claps, P., Clark, M. P., Collins, A. L., Croke, B., Dathe, A., David, P. C., de Barros, F. P. J., de Rooij, G., Di Baldassarre, G., Driscoll, J. M., Duethmann, D., Dwivedi, R., Eris, E., Farmer, W. H., Feiccabrino, J., Ferguson, G., Ferrari, E., Ferraris, S., Fersch, B., Finger, D., Foglia, L., Fowler, K., Gartsman, B., Gascoin, S., Gaume, E., Gelfan, A., Geris, J., Gharari, S., Gleeson, T., Glendell, M., Gonzalez Bevacqua, A., González-Dugo, M. P., Grimaldi, S., Gupta, A. B., Guse, B., Han, D., Hannah, D., Harpold, A., Haun, S., Heal, K., Helfricht, K., Herrnegger, M., Hipsey, M., Hlaváčiková, H., Hohmann, C., Holko, L., Hopkinson, C., Hrachowitz, M., Illangasekare, T. H., Inam, A., Innocente, C., Istanbulluoglu, E., Jarihani, B., Kalantari, Z., Kalvans, A., Khanal, S., Khatami, S., Kiesel, J., Kirkby, M., Knoben, W., Kochanek, K., Kohnová, S., Kolechkina, A., Krause, S., Kreamer, D., Kreibich, H., Kunstmann, H., Lange, H., Liberato, M. L. R., Lindquist, E., Link, T., Liu, J., Loucks, D. P., Luce, C., Mahé, G., Makarieva, O., Malard, J., Mashtayeva, S., Maskey, S., Mas-Pla, J., Mavrova-Guirguinova, M., Mazzoleni, M., Mernild, S., Misstear, B. D., Montanari, A., Müller-Thomy, H., Nabizadeh, A., Nardi, F., Neale, C., Nesterova, N., Nurtaev, B., Odongo, V. O., Panda, S., Pande, S., Pang, Z., Papacharalampous, G., Perrin, C., Pfister, L., Pimentel, R., Polo, M. J., Post, D., Prieto Sierra, C., Ramos, M.-H., Renner, M., Reynolds, J. E., Ridolfi, E., Rigon, R., Riva, M., Robertson, D. E., Rosso, R., Roy, T., Sá, J. H. M., Salvadori, G., Sandells, M., Schaefli, B., Schumann, A., Scolobig, A., Seibert, J., Servat, E., Shafiei, M., Sharma, A., Sidibe, M., Sidle, R. C., Skaugen, T., Smith, H., Spiessl, S. M., Stein, L., Steinsland, I., Strasser, U., Su, B., Szolgay, J., Tarboton, D., Tauro, F., Thirel, G., Tian, F., Tong, R., Tussupova, K., Tyralis, H., Uijlenhoet, R., van Beek, R., van der Ent, R. J., van der Ploeg, M., Van Loon, A. F., van Meerveld, I., van Nooijen, R., van Oel, P. R., Vidal, J.-P., von Freyberg, J., Vorogushyn, S., Wachniew, P., Wade, A. J., Ward, P., Westerberg, I. K., White, C., Wood, E. F., Woods, R., Xu, Z., Yilmaz, K. K., & Zhang, Y. (2019). Twenty-three unsolved problems in hydrology (UPH): a community perspective. *Hydrological Sciences Journal*, *64*(10), 1141-1158. http://doi.org/10.1080/02626667.2019.1620507.

Collischonn, W., Morelli Tucci, C. E., Clarke, R. T., Dias, P. L. S., & Sampaio de Oliveira, G. (2005). Previsão sazonal de vazão na bacia do rio Uruguai 2: previsão climática-hidrológica. *Revista Brasileira de Recursos Hidricos*, *10*(4), 43-59.

Collischonn, W., Allasia, D., Silva, B. C., & Tucci, C. E. M. (2007). The MGB-IPH model for large-scale rainfall-runoff modelling. *Hydrological Sciences Journal*, *52*(5), 878-895. http://doi.org/10.1623/hysj.52.5.878.

Collischonn, W., Brêda, J. P. L. F., Wongchuig, S., Ruhoff, A., Paiva, R. C. D., Fan, F. M., Machado Filho, R. C., & Ramalho, N. (2024). Unprecedented April-May 2024 rainfall in South Brazil sets new record. *Revista Brasileira de Recursos Hídricos*, *29*, e50. http://doi.org/10.1590/2318-0331.292420240088.

Collischonn, W., Fan, F. M., Possantti, I., Dornelles, F., Paiva, R., Medeiros, M. S., Michel, G. P., Magalhães Filho, F. J. C., Moraes, S. R., Marcuzzo, F. F. N., Michel, R. D. L., Beskow, T. L. C., Beskow, S., Fernandes, E. H. L., Santos, L. L., Ruhoff, A., Kobiyama, M., Collares, G. L., Buffon, F., Duarte, E., Lima, S., Meirelles, F. S. C., & Piccilli, D. G. A. (2025). The exceptional hydrological disaster of April-May 2024 in southern Brazil. *Revista Brasileira de Recursos Hídricos*, *30*, e1. http://doi.org/10.1590/2318-0331.302520240119.

Copernicus Browser. (2025). Retrieved in 2025, February 15, from https://browser.dataspace.copernicus.eu/

Fan, F., Collischonn, W., Meller, A., & Botelho, L. C. M. (2014). Ensemble streamflow forecasting experiments in a tropical basin: the São Francisco river case study. *Journal of Hydrology*, *519*, 2906-2919. http://doi.org/10.1016/j.jhydrol.2014.04.038.

Fan, F. M., Meller, A., & Collischonn, W. (2015). Incorporação de filtro numérico de separação de escoamento na assimilação de dados para previsão de vazões utilizando modelagem hidrológica. *Revista Brasileira de Recursos Hídricos*, *20*(2), 472-483. http://doi.org/10.21168/rbrh.v20n2.p472-483.

Fan, F., Fleischmann, A., Siqueira, V., Nectoux, M., Giacomelli, L., & Collischonn, W. (2019). Development of a flood forecasting system for the State of Rio Grande do Sul, Brazil. *Geophysical Research Abstracts*, *21*, 1.

Farr, T. G., Rosen, P. A., Caro, E., Crippen, R., Duren, R., Hensley, S., Kobrick, M., Paller, M., Rodriguez, E., Roth, L., Seal, D., Shaffer, S., Shimada, J., Umland, J., Werner, M., Oskin, M., Burbank, D., & Alsdorf, D. E. (2007). The shuttle radar topography mission. *Reviews of Geophysics*, *45*(2), 2005RG000183. http://doi.org/10.1029/2005RG000183.

Fleischmann, A., Siqueira, V., Paris, A., Collischonn, W., Paiva, R., Pontes, P., Crétaux, J. F., Bergé-Nguyen, M., Biancamaria, S., Gosset, M., Calmant, S., & Tanimoun, B. (2018). Modelling hydrologic and hydrodynamic processes in basins with large semi-arid wetlands. *Journal of Hydrology*, *561*, 943-959. http://doi.org/10.1016/j.jhydrol.2018.04.041.

Fleischmann, A., Collischonn, W., Paiva, R., & Tucci, C. E. (2019a). Modeling the role of reservoirs versus floodplains on large-scale river hydrodynamics. *Natural Hazards*, *99*(2), 1075-1104. http://doi.org/10.1007/s11069-019-03797-9.

Fleischmann, A., Paiva, R., & Collischonn, W. (2019b). Can regional to continental river hydrodynamic models be locally relevant? A cross-scale comparison. *Journal of Hydrology: X*, *3*, 100027. http://doi.org/10.1016/j.hydroa.2019.100027.

GaúchaZH. (2024a, 2 de maio). Guaíba pode ultrapassar os 5 metros, aponta projeção do Instituto de Pesquisas Hidráulicas da UFRGS. *GZH*, Porto Alegre. Retrieved in 2025, February 15, from https://gauchazh.clicrbs.com.br/ambiente/noticia/2024/05/guaiba-pode-ultrapassar-os-5-metros-aponta-projecao-do-instituto-de-pesquisas-hidraulicas-da-ufrgs-clvplcxi1019d014aes0o62ev.html

GaúchaZH. (2024b, 3 de maio). Guaíba deve atingir 5 metros de cheia neste fim de semana, aponta hidrólogo. *GZH*, Porto Alegre. Retrieved in 2025, February 15, from https://gauchazh.clicrbs.com.br/porto-alegre/noticia/2024/05/guaiba-deve-atingir-5-metros-de-cheia-neste-fim-de-semana-aponta-hidrologo-clvre1x5q02f7011w99qc4po1.html

GaúchaZH. (2024c, 3 de maio). Hidrólogos pedem que Porto Alegre prepare evacuação. *GZH*, Porto Alegre. Retrieved in 2025, February 15, from https://gauchazh.clicrbs.com.br/porto-alegre/noticia/2024/05/hidrologos-pedem-que-porto-alegre-prepare-evacuacao-veja-como-seria-inundacao-clvqsldrv022l011w51s0vgfk.html

GaúchaZH. (2024d, 4 de maio). Baixa da inundação em Porto Alegre deverá ocorrer "vagarosamente", apontam pesquisadores da UFRG. *GZH*, Porto Alegre. Retrieved in 2025, February 15, from https://gauchazh.clicrbs.com.br/porto-alegre/noticia/2024/05/baixa-da-inundacao-em-porto-alegre-devera-ocorrer-vagarosamente-apontam-pesquisadores-da-ufrgs-clvspdeha022u014aeqm0605c.html

GaúchaZH. (2024e, 10 de maio). Quais são as chances de o Guaíba ter um "repique" nos próximos dias. *GZH*, Porto Alegre. Retrieved in 2025, February 15, from https://gauchazh.clicrbs.com.br/porto-alegre/noticia/2024/05/quais-as-chances-de-o-guaiba-ter-um-repique-nos-proximos-dias-clw0fej7i0004019atpu4gc9w.html

GaúchaZH. (2024f, 14 de maio). Hidrólogos dizem que Guaíba deve começar a baixar nesta quarta-feira. *GZH*, Porto Alegre. Retrieved in 2025, February 15, from https://gauchazh.clicrbs.com.br/porto-alegre/noticia/2024/05/hidrologo-diz-que-guaiba-deve-comecar-a-baixar-nesta-quarta-feira-clw6z057t001u01edv4na5x76.html

Germano, A. O., Lamberty, D., Silva, E. D., Buffon, F. T., & Pedrollo, M. C. R. (2024). *Avaliação indireta do nível máximo das águas do delta do Rio Jacuí na região central de Porto Alegre, entre as estações Cais Mauá C6 e Usina do Gasômetro, na grande cheia de maio de 2024*. Brasília: Serviço Geológico do Brasil. Retrieved in 2025, February 15, from https://rigeo.sgb.gov.br/handle/doc/24911

Kuller, M., Schoenholzer, K., & Lienert, J. (2021). Creating effective flood warnings: a framework from a critical review. *Journal of Hydrology*, *602*, 126708. http://doi.org/10.1016/j.jhydrol.2021.126708.

Lopes, V. A. R., Fan, F. M., Pontes, P. R. M., Siqueira, V. A., Collischonn, W., & Motta Marques, D. (2018). A first integrated modelling of a river-lagoon large-scale hydrological system for forecasting purposes. *Journal of Hydrology*, *565*, 177-196. http://doi.org/10.1016/j.jhydrol.2018.08.011.

Medeiros, M. S. (2025). *A inundação sem precedentes de 2024 no sul do Brasil vivenciada como um previsor hidrológico*. S.l.: s.n.

Meller, A., Bravo, J. M., & Collischonn, W. (2012). Assimilação de dados de vazão na previsão de cheias em tempo-real com o modelo hidrológico MGB-IPH. *Revista Brasileira de Recursos Hídricos*, *17*(3), 209-224. http://doi.org/10.21168/rbrh.v17n3.p209-224.

Möller, O. O., & Lorenzzentti, J. A. (1996). The Patos Lagoon summertime circulation and dynamics. *Continental Shelf Research*, *16*(3), 335-351. http://doi.org/10.1016/0278-4343(95)00014-R.

Moller, M., Castaing, P., Salomon, J.-C., & Lazure, P. (2001). The influence of local and non-local forcing effects on the subtidal circulation of Patos Lagoon. *Estuaries*, *24*(2), 297-311. http://doi.org/10.2307/1352953.

Montanari, A., Merz, B., & Blöschl, G. (2024). HESS Opinions: the sword of Damocles of the impossible flood. *Hydrology and Earth System Sciences*, *28*(12), 2603-2615. http://doi.org/10.5194/hess-28-2603-2024.

Najafi, H., Shrestha, P. K., Rakovec, O., Apel, H., Vorogushyn, S., Kumar, R., Thober, S., Merz, B., & Samaniego, L. (2024). High-resolution impact-based early warning system for riverine flooding. *Nature Communications*, *15*(1), 3726. http://doi.org/10.1038/s41467-024-48065-y.

Paiva, R. C. D., Collischonn, W., & Tucci, C. E. M. (2011). Large scale hydrologic and hydrodynamic modeling using limited data and a GIS based approach. *Journal of Hydrology*, *406*(3-4), 170-181. http://doi.org/10.1016/j.jhydrol.2011.06.007.

Paiva, R. C. D., Collischonn, W., & Buarque, D. C. (2013). Validation of a full hydrodynamic model for large-scale hydrologic modelling in the Amazon. *Hydrological Processes*, *27*(3), 333-346. http://doi.org/10.1002/hyp.8425.

Pappenberger, F., Thielen, J., & Del Medico, M. (2011). The impact of weather forecast improvements on large scale hydrology: analysing a decade of forecasts of the European Flood Alert System. *Hydrological Processes*, *25*(7), 1091-1113. http://doi.org/10.1002/hyp.7772.

Paz, A. R., Collischonn, W., Tucci, C. E. M., & Padovani, C. R. (2011). Large-scale modelling of channel flow and floodplain inundation dynamics and its application to the Pantanal (Brazil). *Hydrological Processes*, *25*(9), 1498-1516. http://doi.org/10.1002/hyp.7926.

Pechlivanidis, I. G., Du, Y., Bennett, J., Boucher, M.-A., Chang, A. Y. Y., Crochemore, L., Dasgupta, A., Di Baldassarre, G., Luterbacher, J., Pappenberger, F., Ramos, M.-H., Slater, L., Uhlenbrook, S., Wetterhall, F., Wood, A., Lavado-Casimiro, W., Yoshimura, K., Imhoff, R., van Oevelen, P. J., Cantone, C., Cattoën, C., Pimentel, R., & Werner, M. (2025). Enhancing research-to-operations in hydrological forecasting: innovations across scales and horizons. *Bulletin of the American Meteorological Society*, *106*(5), E894-E919. http://doi.org/10.1175/BAMS-D-24-0322.1.

Pontes, P. R. M., Collischonn, W., Fan, F. M., Paiva, R. C. D., & Buarque, D. C. (2015). Modelagem hidrológica e hidráulica de grande escala com propagação inercial de vazões. *Revista Brasileira de Recursos Hídricos*, *20*(4), 888-904. http://doi.org/10.21168/rbrh.v20n4.p888-904.

Possa, T. M., Collischonn, W., Jardim, P. F., & Fan, F. M. (2022). Hydrological-hydrodynamic simulation and analysis of the possible influence of the wind in the extraordinary flood of 1941 in Porto Alegre. *Revista Brasileira de Recursos Hídricos*, *27*, e29. http://doi.org/10.1590/2318-0331.272220220028.

Rolim, A., Paz, D. A., Collischonn, W., Tucci, C. E. M., Clarke, R. T., & Allasia, D. (2007). *Data assimilation in a large-scale distributed hydrological model for medium-range flow forecasts* (IAHS Publ., No. 313). Wallingford: IAHS Press.

Silveira, A., Dornelles, F., Avruch Goldenfum, J., Possa, T., Collischonn, W., & Royer Moraes, S. (2023). A histórica cheia de 1941 na bacia hidrográfica do Guaíba. In F. Zanandrea, M. Kobiyama, G. P. Michel, A. S. Fleischmann, & W. Collischonn (Eds.), *Desastres e água: eventos históricos no Brasil*. Porto Alegre: ABRHidro.

Siqueira, V. A., Collischonn, W., Fan, F. M., & Chou, S. C. (2016). Previsão de cheias por conjunto a partir de previsões operacionais do EPS regional Eta na bacia do Taquari-Antas/RS. *Revista Brasileira de Recursos Hídricos*, *21*(3), 587-602. http://doi.org/10.1590/2318-0331.011616004.

Siqueira, V. A., Paiva, R. C. D., Fleischmann, A. S., Fan, F. M., Ruhoff, A. L., Pontes, P. R. M., Paris, A., Calmant, S., & Collischonn, W. (2018). Toward continental hydrologic-hydrodynamic modeling in South America. *Hydrology and Earth System Sciences*, *22*(9), 4815-4842. http://doi.org/10.5194/hess-22-4815-2018.

Siqueira, V. A., Fan, F. M., Paiva, R. C. D., Ramos, M. H., & Collischonn, W. (2020). Potential skill of continental-scale, medium-range ensemble streamflow forecasts for flood prediction in South America. *Journal of Hydrology*, *590*, 125430. http://doi.org/10.1016/j.jhydrol.2020.125430.

Universidade Federal do Rio Grande do Sul – UFRGS. (2025a). *Notícias IPH*. Retrieved in 2025, February 15, from https://www.ufrgs.br/iph/noticias/

Universidade Federal do Rio Grande do Sul – UFRGS. (2025b). *Cheias no Rio Grande do Sul*. Retrieved in 2025, February 15, from https://storymaps.arcgis.com/stories/a81d69f4bccf42989609e3fe64d8ef48https://www.ufrgs.br/iph/noticias/

U.S. Army Corps of Engineers – USACE. (2021). *HEC-RAS River Analysis System user's manual (version 6.0.)*. New York.

Zhang, L., Song, Y., Cui, H., Lu, M., Li, C., Yuan, B., Wang, B., Lall, U., & Yang, J. (2025). Foundation models as assistive tools in hydrometeorology: opportunities, challenges, and perspectives. *Water Resources Research*, *61*(4), e2024WR039553. http://doi.org/10.1029/2024WR039553.

Authors contributions

Matheus Sampaio Medeiros: Conceptualization, methodology, data analysis, writing, review, and editing.

Fernando Mainardi Fan: Conceptualization, methodology, data analysis, writing, review, and editing.

Rodrigo Cauduro Dias de Paiva: Conceptualization, methodology, data analysis, writing and review.

Editor-in-Chief: Adilson Pinheiro

Associated Editor: Mariana Madruga de Brito